Assembly Question

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COMETFWC052

Question

Match the logic gates in Column A with their equivalents in Column B.

Column-A	Column-B
1. AND GATE	P. XOR GATE
2. OR GATE	Q. XNOR GATE
3. NOT GATE	R. NAND GATE
	S. NOR GATE

TABLE 11: Table-1

- a) P-2, Q-4, R-1, S-3
- b) P-4, Q-2, R-1, S-3
- c) P-2, Q-4, R-3, S-1
- d) P-4, Q-2, R-3, S-1

Solution

Using logic identities:

$$A \cdot B = (A\ NAND\ B)'$$

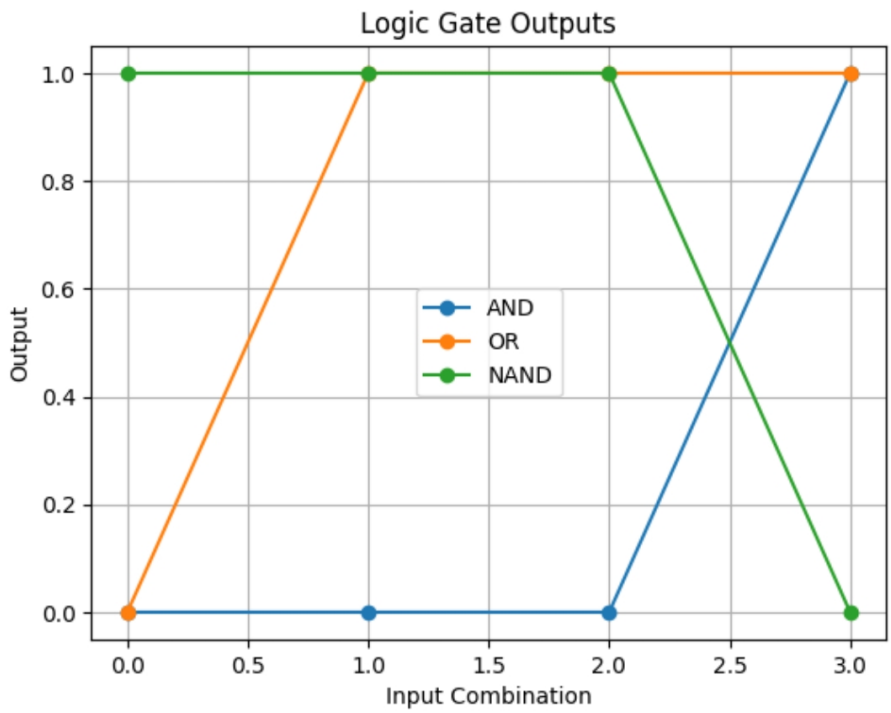
$$A + B = (A\ NOR\ B)'$$

$$A' = A\ XNOR\ 0$$

Thus the correct matching is

$$P - 2,\ Q - 4,\ R - 1,\ S - 3$$

Hence the correct option is

a) P-2, Q-4, R-1, S-3

Graph Output